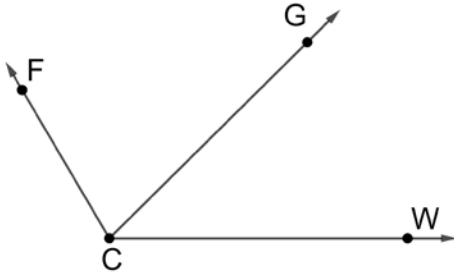


Angle FCW is shown. $m\angle FCW = 120^\circ$, $m\angle FCG = (3x + 4)^\circ$ and $m\angle GCW = (2x - 1)^\circ$.



1. Find the value of x .

$$\begin{aligned} 3x + 4 + 2x - 1 &= 120 \\ 5x + 3 &= 120 \\ 5x &= 117 \\ x &= 23.4 \end{aligned}$$

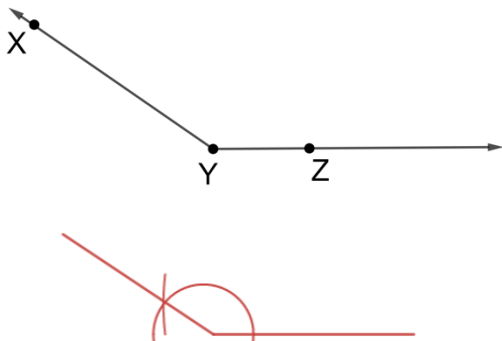
2. Find $m\angle FCG$.

$$3 \cdot 23.4 + 4 = 74.2$$

3. Find $m\angle GCW$.

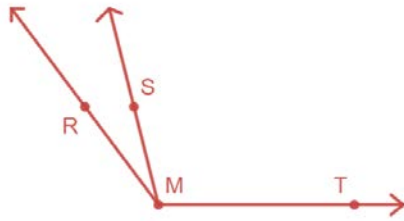
$$2 \cdot 23.4 - 1 = 45.8$$

4. Use a straightedge and a compass to copy $\angle XYZ$.



Given that ray MS lies inside $\angle RMT$ such that $m\angle RMT = (10x)^\circ$, $m\angle RMS = (2x - 5)^\circ$ and $m\angle SMT = (-2x + 145)^\circ$.

5. Draw a diagram that represents $\angle RMT$ and $\overrightarrow{MS}$.



6. Find the value of x .

$$\begin{aligned}10x &= 2x - 5 - 2x + 145 \\10x &= 140 \\x &= 14\end{aligned}$$

7. Find $m\angle RMT$.

$$m\angle RMT = 10(14) = 140^\circ$$

8. Find $m\angle RMS$.

$$m\angle RMS = 2(14) - 5 = 23^\circ$$

9. Find $m\angle SMT$.

$$m\angle SMT = -2(14) + 145 = 117^\circ$$

10. $\overrightarrow{DN}$ lies inside $\angle LDP$. If $\angle LDP$ is a straight angle, and $m\angle LDN$ is five times $m\angle NDP$, find $m\angle LDN$.

$$\begin{aligned}m\angle LDN + m\angle NDP &= 180 \\5x + x &= 180 \\6x &= 180 \\x &= 30 \\m\angle LDN &= 5(30) = 150^\circ\end{aligned}$$